The Parameterized Complexity of Learning Monadic Second-Order Logic

Nina Runde
(Steffen van Bergerem, Martin Grohe)

February 14, 2024

RWTH Aachen University

Learning Problems

Related Problems

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

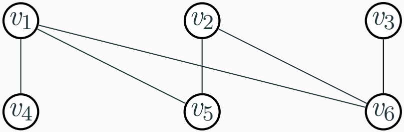
Related Problems

Related Problems

Model Checking

Model Checking

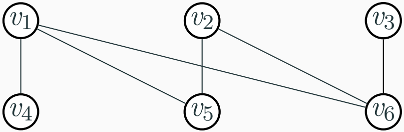
Input: Graph



Related Problems

Model Checking

Input: Graph



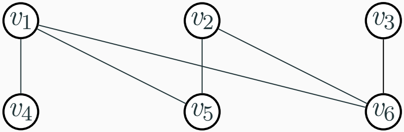
MSO-formula

$\psi_{\text{bipartite}} = \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg \big(Z(z_1) \leftrightarrow Z(z_2) \big) \Big)$

Related Problems

Model Checking

Input: Graph



MSO-formula

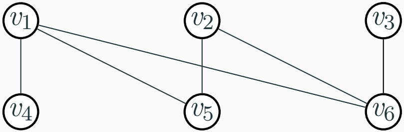
$\psi_{\text{bipartite}} = \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg\big(Z(z_1) \leftrightarrow Z(z_2)\big)\Big)$

Solution:

Related Problems

Model Checking

Input: Graph



MSO-formula

$\psi_{\text{bipartite}} = \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg\big(Z(z_1) \leftrightarrow Z(z_2)\big)\Big)$

Solution:

Yes! $G \models \psi_{\text{bipartite}}$

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Enumeration $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, formula $\varphi(x_1, x_2)$ from $\mathbb{L}$

Problem: Enumerate all $(v_1, v_2) \in V(G)^2$ such that $G \models \varphi(v_1, v_2)$

Related Problems

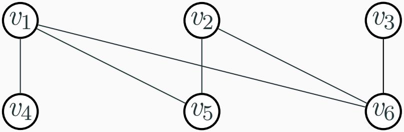
Related Problems

Enumeration

Related Problems

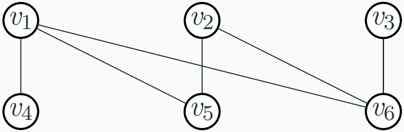
Enumeration

Input: Graph



ENUMERATION

Input: Graph

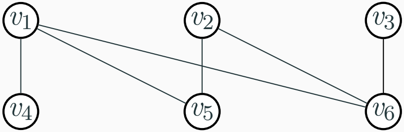


MSO-formula

$$\psi_{\text{bipartite}}(x_1, x_2)$$
$$= \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg\big(Z(z_1) \leftrightarrow Z(z_2)\big)\Big) \wedge Z(x_1) \wedge Z(x_2)$$

ENUMERATION

Input: Graph



MSO-formula

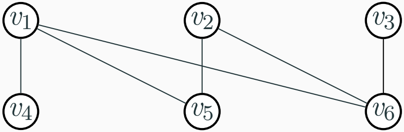
$\psi_{\text{bipartite}}(x_1, x_2)$
$= \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg \big(Z(z_1) \leftrightarrow Z(z_2) \big) \Big) \wedge Z(x_1) \wedge Z(x_2)$

Solution:

Related Problems

Enumeration

Input: Graph



MSO-formula

$\psi_{\text{bipartite}}(x_1, x_2)$
$= \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg \big(Z(z_1) \leftrightarrow Z(z_2) \big) \Big) \wedge Z(x_1) \wedge Z(x_2)$

Solution:

$(v_1, v_2), (v_1, v_3), (v_2, v_3), (v_4, v_5), (v_4, v_6), (v_5, v_6), (v_2, v_1) \ldots$

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Enumeration $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, formula $\varphi(x_1, x_2)$ from $\mathbb{L}$

Problem: Enumerate all $(v_1, v_2) \in V(G)^2$ such that $G \models \varphi(v_1, v_2)$

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Enumeration $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, formula $\varphi(x_1, x_2)$ from $\mathbb{L}$

Problem: Enumerate all $(v_1, v_2) \in V(G)^2$ such that $G \models \varphi(v_1, v_2)$

Consistent Learning $(\mathbb{G}, \mathbb{L})$

Input: Graph G

Problem: Find φ from $\mathbb{L}$

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Enumeration $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, formula $\varphi(x_1, x_2)$ from $\mathbb{L}$

Problem: Enumerate all $(v_1, v_2) \in V(G)^2$ such that $G \models \varphi(v_1, v_2)$

Consistent Learning $(\mathbb{G}, \mathbb{L})$

Input: Graph G , $\mathcal{S} = \{((v_1, v_2), +), ((v_3, v_4), -)\}$

Problem: Find φ from $\mathbb{L}$

Related Problems

Model Checking $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, sentence φ from $\mathbb{L}$

Problem: $G \models \varphi$?

Enumeration $(\mathbb{G}, \mathbb{L})$

Input: Graph $G \in \mathbb{G}$, formula $\varphi(x_1, x_2)$ from $\mathbb{L}$

Problem: Enumerate all $(v_1, v_2) \in V(G)^2$ such that $G \models \varphi(v_1, v_2)$

Consistent Learning $(\mathbb{G}, \mathbb{L})$

Input: Graph G , $\mathcal{S} = \{((v_1, v_2), +), ((v_3, v_4), -)\}$

Problem: Find φ from $\mathbb{L}$ such that $G \models \varphi(v_1, v_2)$ and $G \not\models \varphi(v_3, v_4)$

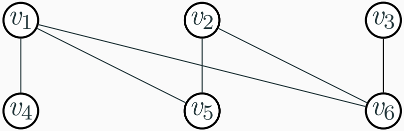
Related Problems

Related Problems

Consistent-Learning

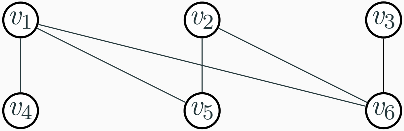
CONSISTENT-LEARNING

Input: Graph



CONSISTENT-LEARNING

Input: Graph

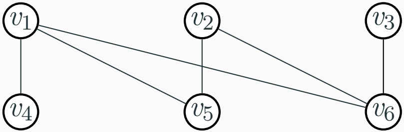


$$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$$

Related Problems

CONSISTENT-LEARNING

Input: Graph

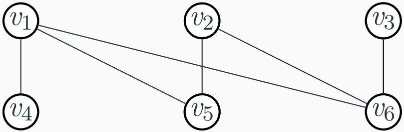


$$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$$

Solution:

Consistent-Learning

Input: Graph

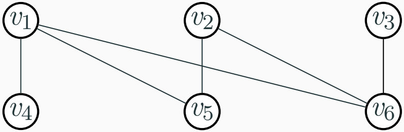


$$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$$

Solution: MSO-formula

CONSISTENT-LEARNING

Input: Graph



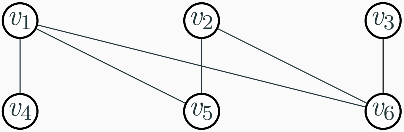
$$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$$

Solution: MSO-formula

$$\psi_{\text{bipartite}}(x_1)$$
$$= \exists Z \forall z_1 \forall z_2 \Big(E(z_1, z_2) \to \neg \big(Z(z_1) \leftrightarrow Z(z_2) \big) \Big) \land Z(x_1) \land Z(v_2)$$

CONSISTENT-LEARNING

Input: Graph



$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$

Solution: MSO-formula

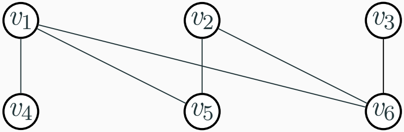
$\psi\prime(x_1)$

$= (x_1 = v_1) \vee (x_1 = v_3)$

Related Problems

Consistent-Learning

Input: Graph



$\mathcal{S} = \{((v_1), +), ((v_3), +), ((v_4), -), ((v_5), -)\}$

Solution: MSO-formula

$\psi_{\prime}(x_1)$

$= (x_1 = v_1) \vee (x_1 = v_3)$

Overfitting (Limit parameter variables and quantifier rank)

Problem Definition MSO-Consistent-Learn

Problem Definition MSO-Consistent-Learn

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Problem Definition MSO-Consistent-Learn

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Solution: $h_{\varphi,\bar{v}}$

Problem Definition MSO-Consistent-Learn

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Solution: $h_{\varphi,\bar{v}}$

$$\varphi(\bar{x}, \bar{y}) \in \mathsf{MSO}[\Lambda \cup \{E\}, q, k + \ell]$$

-

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Solution: $h_{\varphi,\bar{v}}$

- $\varphi(\bar{x}, \bar{y}) \in \mathsf{MSO}[\Lambda \cup \{E\}, q, k + \ell]$

  instance variables parameter variables

Problem Definition MSO-CONSISTENT-LEARN

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Solution: $h_{\varphi,\bar{v}}$

- $\varphi(\bar{x}, \bar{y}) \in \mathsf{MSO}[\Lambda \cup \{E\}, q, k + \ell]$

  instance variables   parameter variables
- parameter setting $\bar{v} \in V(G)^\ell$

Problem Definition MSO-Consistent-Learn

Input:

- Λ-labeled graph G
- $k, \ell, q \in \mathbb{N}$
- training set $\mathcal{S} \subseteq V(G)^k \times \{+, -\}$

Solution: $h_{\varphi,\bar{v}}$

- $\varphi(\bar{x}, \bar{y}) \in \mathsf{MSO}[\Lambda \cup \{E\}, q, k + \ell]$

  instance variables parameter variables

- parameter setting $\bar{v} \in V(G)^\ell$
- consistent

  $(\bar{w}_1, +) \in \mathcal{S}$ then $G \models \varphi(\bar{w}_1, \bar{v})$

  $(\bar{w}_2, -) \in \mathcal{S}$ then $G \not\models \varphi(\bar{w}_2, \bar{v})$

Parameterized Complexity

Parameter: $|\tau| + k + q + \ell$

Parameterized Complexity

Parameter: $|\tau| + k + q + \ell$

fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Parameter: $|\tau| + k + q + \ell$
fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Finding the Formula:

Parameterized Complexity

Parameter: $|\tau| + k + q + \ell$
fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Finding the Formula:

- The set MSO$[\tau, q, k + \ell]$ is finite up to equivalence

Parameterized Complexity

Parameter: $|\tau| + k + q + \ell$

fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Finding the Formula:

- The set MSO$[\tau, q, k + \ell]$ is finite up to equivalence
- we can fix a normal form and iterate through all formulas

Parameter: $|\tau| + k + q + \ell$
fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Finding the Formula:

- The set MSO$[\tau, q, k + \ell]$ is finite up to equivalence
- we can fix a normal form and iterate through all formulas

The Problem:

Parameter: $|\tau| + k + q + \ell$

fixed-parameter tractable: $f(|\tau|, k, q, \ell) \cdot p(|G|, |\mathcal{S}|)$

Finding the Formula:

- The set MSO$[\tau, q, k + \ell]$ is finite up to equivalence
- we can fix a normal form and iterate through all formulas

The Problem:

Given φ, find parameter setting $\bar{v}$

Related Work

Previous results

Previous results

First-Order

Previous results

First-Order

- FPT on classes of graphs that are nowhere-dense
- AW[*]-hard on classes of graphs that are not nowhere-dense
  (van Bergerem, Grohe, Ritzert, 2021)

Previous results

First-Order

- FPT on classes of graphs that are nowhere-dense
- AW[*]-hard on classes of graphs that are not nowhere-dense
  (van Bergerem, Grohe, Ritzert, 2021)

Monadic Second-Order

Previous results

First-Order

- FPT on classes of graphs that are nowhere-dense
- AW[*]-hard on classes of graphs that are not nowhere-dense
  (van Bergerem, Grohe, Ritzert, 2021)

Monadic Second-Order

- MSO-Consistent-Learn for $k = 1$ on strings
  (Grohe, Löding, Ritzert, 2017)
- MSO-Consistent-Learn for $k = 1$ on trees
  (Grienenberger, Ritzert, 2019)

The One-Dimensional Case

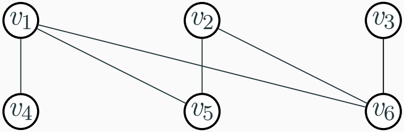
One-Dimensional Training Data

One-Dimensional Training Data

Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$

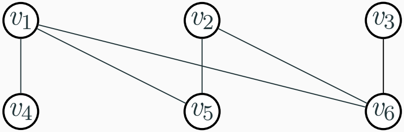
One-Dimensional Training Data

Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$



One-Dimensional Training Data

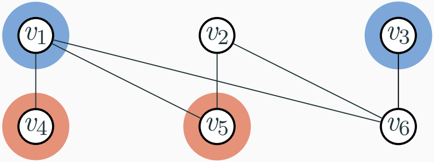
Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$



Add two unary relations P and N

One-Dimensional Training Data

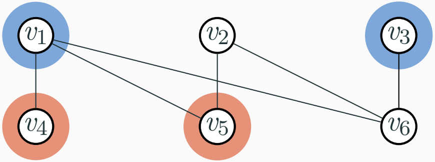
Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$



Add two unary relations P and N

One-Dimensional Training Data

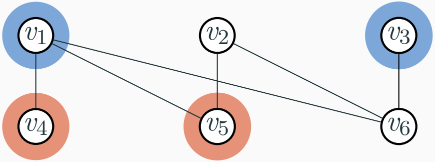
Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$



Add two unary relations P and N
with $v \in P$ iff $(v, +) \in \mathcal{S}$ and $v \in N$ iff $(v, -) \in \mathcal{S}$

One-Dimensional Training Data

Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$

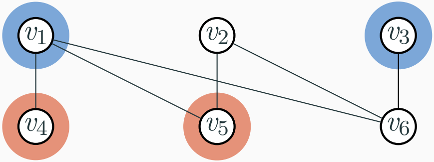


Add two unary relations P and N
with $v \in P$ iff $(v, +) \in \mathcal{S}$ and $v \in N$ iff $(v, -) \in \mathcal{S}$

Choose formula

One-Dimensional Training Data

Given training set $\mathcal{S} = \{(v_1, +), (v_3, +), (v_4, -), (v_5, -)\}$, graph G and formula $\varphi(x, \bar{y})$



Add two unary relations P and N with $v \in P$ iff $(v, +) \in \mathcal{S}$ and $v \in N$ iff $(v, -) \in \mathcal{S}$

Choose formula

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$

Solve Parameter Learning Using Model Checking

Solve Parameter Learning Using Model Checking

$$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$$

Solve Parameter Learning Using Model Checking

$$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$$

model-checking $\varphi_3 = \exists y_1, \ldots \exists y_\ell \; \varphi_2(y_1, \ldots y_\ell)$

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \ldots \exists y_\ell \; \varphi_2(y_1, \ldots y_\ell)$

- G' with $X_1 = \{v_1\}$

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \ldots \exists y_\ell \; \varphi_2(y_1, \ldots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \wedge X_1(y_1)$

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \land N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \dots \exists y_\ell \; \varphi_2(y_1, \dots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \land X_1(y_1)$
- model-checking

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \ldots \exists y_\ell \; \varphi_2(y_1, \ldots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \wedge X_1(y_1)$
- model-checking

ℓ

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \land N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \dots \exists y_\ell \; \varphi_2(y_1, \dots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \land X_1(y_1)$
- model-checking

$\ell \cdot n$

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \land N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \dots \exists y_\ell \; \varphi_2(y_1, \dots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \land X_1(y_1)$
- model-checking

$\ell \cdot n \cdot$ time(MC)

Solve Parameter Learning Using Model Checking

$\varphi_2(\bar{y}) = \forall x P(x) \to \varphi(x, \bar{y}) \wedge N(x) \to \neg\varphi(x, \bar{y})$

model-checking $\varphi_3 = \exists y_1, \dots \exists y_\ell \; \varphi_2(y_1, \dots y_\ell)$

- G' with $X_1 = \{v_1\}$
- $\varphi_4 = \varphi_3 \wedge X_1(y_1)$
- model-checking

$\ell \cdot n \cdot$ time(MC)

If MSO-Model-Checking($\mathbb{G}$) is fixed-parameter tractable, so is one-dimensional MSO-Consistent-Learning($\mathbb{G}$).

Hardness

Hardness

- FO-Model-Checking $\leq_{\text{fpt}}$ FO-Consistent-Learning

Hardness

- FO-Model-Checking $\leq_{\text{fpt}}$ FO-Consistent-Learning
- FO-Model-Checking $\leq_{\text{fpt}}$ MSO-Consistent-Learning

Hardness

- FO-Model-Checking $\leq_{\text{fpt}}$ FO-Consistent-Learning
- FO-Model-Checking $\leq_{\text{fpt}}$ MSO-Consistent-Learning

- FO-Model-Checking is AW[*]-hard

Hardness

- FO-Model-Checking $\leq_{\mathrm{fpt}}$ FO-Consistent-Learning
- FO-Model-Checking $\leq_{\mathrm{fpt}}$ MSO-Consistent-Learning

- FO-Model-Checking is AW[*]-hard
- MSO-Model-Checking is para-NP-hard

Hardness

- FO-Model-Checking $\leq_{\text{fpt}}$ FO-Consistent-Learning
- FO-Model-Checking $\leq_{\text{fpt}}$ MSO-Consistent-Learning

- FO-Model-Checking is AW[*]-hard
- MSO-Model-Checking is para-NP-hard

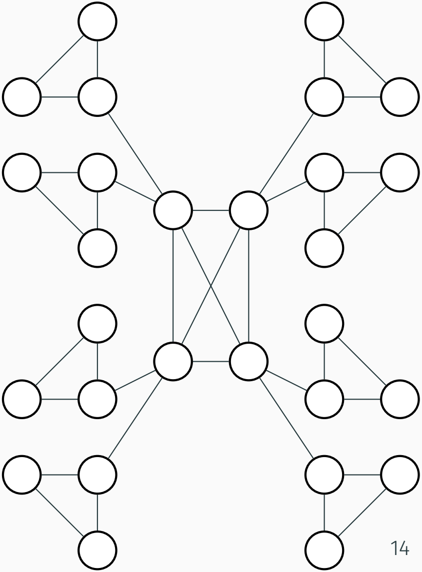
- MSO-Model-Checking $\leq_{\text{fpt}}$ MSO-Consistent-Learning

Type

Type

Example: $\exists x\ \psi(x)$

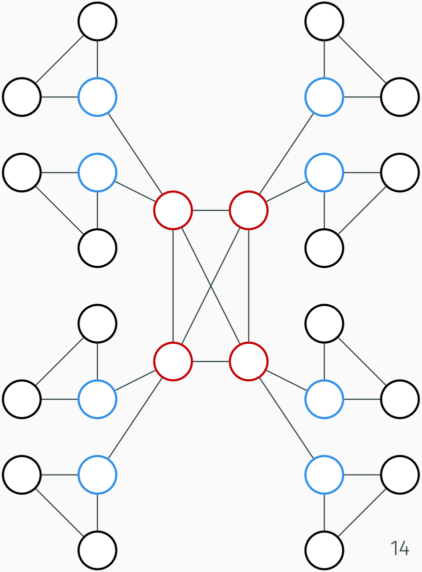
Example: $\exists x\ \psi(x)$



Type

Example: $\exists x \; \psi(x)$

$tp_{q,\tau}(G, v) = \{\varphi(x) \mid G \models \varphi(v)\}$
for $\varphi \in$ MSO$[\tau, q, 1]$

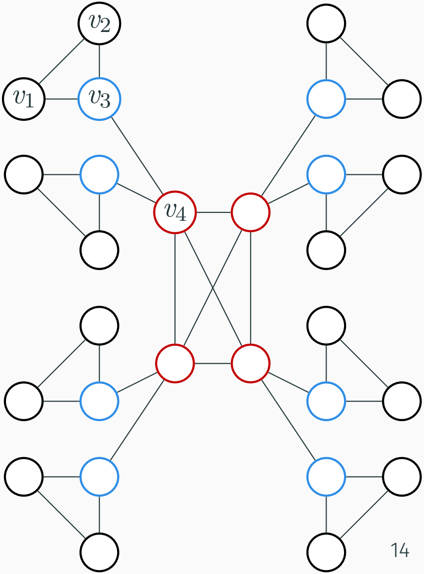


Type

Example: $\exists x\ \psi(x)$

$tp_{q,\tau}(G, v) = \{\varphi(x) \mid G \models \varphi(v)\}$
for $\varphi \in$ MSO$[\tau, q, 1]$

Set of representatives:
$R = \{v_1, v_3, v_4\}$



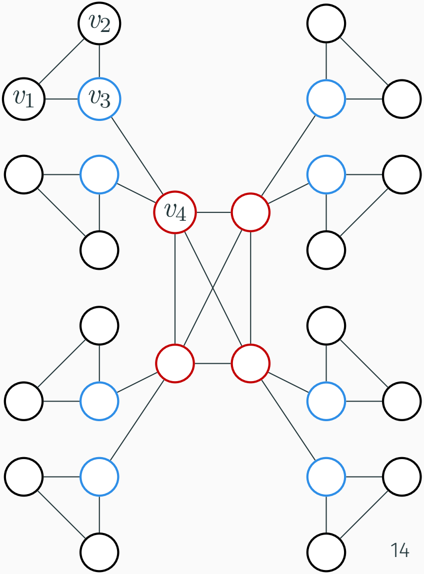
Type

Example: $\exists x\ \psi(x)$

$tp_{q,\tau}(G, v) = \{\varphi(x) \mid G \models \varphi(v)\}$
for $\varphi \in \mathsf{MSO}[\tau, q, 1]$

Set of representatives:
$R = \{v_1, v_3, v_4\}$

Run Consistent-Learn
oracle on G
with $\mathcal{S} = \{(v_1, +), (v_2, -)\}$



Type

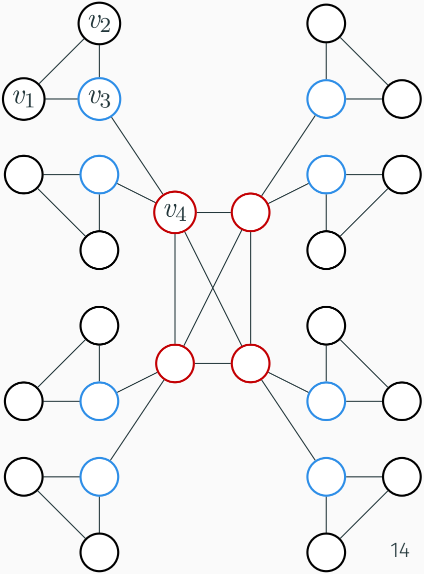
Example: $\exists x\ \psi(x)$

$tp_{q,\tau}(G, v) = \{\varphi(x) \mid G \models \varphi(v)\}$
for $\varphi \in$ MSO$[\tau, q, 1]$

Set of representatives:
$R = \{v_1, v_3, v_4\}$

Run Consistent-Learn
oracle on G
with $\mathcal{S} = \{(v_1, +), (v_2, -)\}$

What about $\exists X\ \psi(X)$?



Results 1-Dimensional Case

Results 1-Dimensional Case

- On graphs of bounded clique-width, one-dimensional MSO-Consistent-Learn is fixed-parameter tractable

Results 1-Dimensional Case

- On graphs of bounded clique-width, one-dimensional MSO-Consistent-Learn is fixed-parameter tractable
- On arbitrary graphs, one dimensional MSO-Consistent-Learn is para-NP-hard

The Higher-Dimensional Case

The Higher-Dimensional Case

The Problem

The Problem

- We can't encode the examples using relations.

The Problem

- We can't encode the examples using relations.
- k-ary relations may destroy the structure and the tractability of the model-checking problem

The Problem

- We can’t encode the examples using relations.
- k-ary relations may destroy the structure and the tractability of the model-checking problem
- Unary relations increase the size of the vocabulary to much

The Higher-Dimensional Case

The Problem

- We can't encode the examples using relations.
- k-ary relations may destroy the structure and the tractability of the model-checking problem
- Unary relations increase the size of the vocabulary to much (one for each example and position)

Probably Approximately Correct Learning
(Valian, 1984)

PAC-Learning

Probably Approximately Correct Learning
(Valian, 1984)

Extra assumptions:

- Examples taken from some data generating distribution
- Find hypothesis that has a small error with high probability

PAC-Learning

Probably Approximately Correct Learning
(Valian, 1984)

Extra assumptions:

- Examples taken from some data generating distribution
- Find hypothesis that has a small error with high probability

Number of examples

PAC-Learning

Probably Approximately Correct Learning
(Valian, 1984)

Extra assumptions:

- Examples taken from some data generating distribution
- Find hypothesis that has a small error with high probability

Number of examples $\leftarrow$ VC dimension

PAC-Learning

Probably Approximately Correct Learning
(Valian, 1984)

Extra assumptions:

- Examples taken from some data generating distribution
- Find hypothesis that has a small error with high probability

Number of examples $\leftarrow$ VC dimension $\leftarrow$ Parameterization

Probably Approximately Correct Learning
(Valian, 1984)

Extra assumptions:

- Examples taken from some data generating distribution
- Find hypothesis that has a small error with high probability

Number of examples $\leftarrow$ VC dimension $\leftarrow$ Parameterization

But what if we actually want to be consistent?

Consistent Learning in Higher Dimensions

On graphs of bounded clique-width:

Consistent Learning in Higher Dimensions

On graphs of bounded clique-width:

- $\mathcal{O}(g(c, |\Lambda|, q, k, \ell) \cdot |V(G)|^{\ell+1} \cdot m)$

Consistent Learning in Higher Dimensions

On graphs of bounded clique-width:

- $\mathcal{O}(g(c, |\Lambda|, q, k, \ell) \cdot |V(G)|^{\ell+1} \cdot m)$
- $\mathcal{O}\big((m+1)^{g(c,|\Lambda|,q,k,\ell)}|V(G)|^2\big)$

Consistent Learning in Higher Dimensions

On graphs of bounded clique-width:

- $\mathcal{O}(g(c, |\Lambda|, q, k, \ell) \cdot |V(G)|^{\ell+1} \cdot m)$
- $\mathcal{O}\big((m+1)^{g(c,|\Lambda|,q,k,\ell)}|V(G)|^2\big)$
- cannot be improved into an fpt-result

Results

Results

Results

Problem	Dim	Structure	Result

Results

Problem	Dim	Structure	Result
CONSISTENT	$k = 1$	bounded cw	fpt

Results

Problem	Dim	Structure	Result
Consistent	$k = 1$	bounded cw	fpt
Consistent	$k = 1$	arbitrary	para-NP-hard

Results

Problem	Dim	Structure	Result
CONSISTENT	$k = 1$	bounded cw	fpt
CONSISTENT	$k = 1$	arbitrary	para-NP-hard
CONSISTENT	$k > 1$	bounded cw	$\mathcal{O}\big((m+1)^{g(c,\lvert\Lambda\rvert,q,k,\ell)}\lvert V(G)\rvert^2\big)$

Results

Problem	Dim	Structure	Result
Consistent	$k = 1$	bounded cw	fpt
Consistent	$k = 1$	arbitrary	para-NP-hard
Consistent	$k > 1$	bounded cw	$\mathcal{O}\big((m+1)^{g(c,\lvert\Lambda\rvert,q,k,\ell)}\lvert V(G)\rvert^2\big)$
PAC-Learn	$k \geq 1$	bounded cw	fpt